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Erythrocytic Pyruvate Kinase Mutations Causing Hemolytic Anemia, Osteosclerosis, and Secondary Hemochromatosis in Dogs

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Abstract

Background

Erythrocytic pyruvate kinase (PK) deficiency, first documented in Basenjis, is the most common inherited erythroenzymopathy in dogs.

Objectives

To report 3 new breed-specific *PK-LR* gene mutations and a retrospective survey of PK mutations in a small and selected group of Beagles and West Highland White Terriers (WHWT).

Animals

Labrador Retrievers (2 siblings, 5 unrelated), Pugs (2 siblings, 1 unrelated), Beagles (39 anemic, 29 other), WHWTs (22 anemic, 226 nonanemic), Cairn Terrier (n = 1).

Methods

Exons of the *PK-LR* gene were sequenced from genomic DNA of young dogs (<2 years) with persistent highly regenerative hemolytic anemia.

Results

A nonsense mutation (c.799C>T) resulting in a premature stop codon was identified in anemic Labrador Retriever siblings that had osteosclerosis, high serum ferritin concentrations, and severe hepatic secondary hemochromatosis. Anemic Pug and Beagle revealed 2 different missense mutations (c.848T>C, c.994G>A, respectively) resulting in intolerable amino acid changes to protein structure and enzyme function. Breed-specific mutation tests were developed. Among the biased group of 248 WHWTs, 9% and 35% were homozygous (affected) and heterozygous, respectively, for the previously described mutation (mutant allele frequency 0.26). A PK-deficient Cairn Terrier had the same insertion mutation as the affected WHWTs. Of the selected group of 68 Beagles, 35% were PK-deficient and 3% were carriers (0.37).

Conclusions and Clinical Importance

Erythrocytic PK deficiency is caused by different mutations in different dog breeds and causes chronic severe hemolytic anemia, hemosiderosis, and secondary hemochromatosis because of chronic hemolysis and, an as yet unexplained osteosclerosis. The newly developed breed-specific mutation assays simplify the diagnosis of PK deficiency.

Keywords: Genetics, Hereditary Disorders, Metabolism, Reticulocytes

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The enzyme R-type pyruvate kinase (R-PK, E.C 2.7.1.40) is specifically expressed in erythrocytes and is a key regulatory enzyme in anaerobic glycolysis and energy generation.[1,2] Erythrocytic PK deficiency is the most common erythroenzymopathy in dogs[2–12] (Online Mendelian Inheritance in Animals,[13] OMIA 000844-9615), cats[14–18] (OMIA 000844-9685) and humans[19–28] (Online Mendelian Inheritance in Man, OMIM 266200), is inherited as an autosomal recessive trait, and causes a severe chronic or intermittent hemolytic anemia.[2,6–9,11,12] Erythrocytic PK deficiency has been reported in several breeds of dogs

including Basenjis,[3–6] Beagles,[7,8] West Highland White Terriers (WHWT),[9–11] and a Cairn Terrier[12] and is interestingly also associated with progressive osteosclerosis,[3,9] varied hepatopathy,[2,9,12] and an anomalous and dysfunctional M_2-PK expression in erythrocytes.[2,3,8,10]

The *PK-LR* gene (Gene ID: 490425), located on canine chromosome 7, is tissue-specifically spliced into R-PK in late erythroid precursors and L-PK in hepatocytes.[29] A single base pair (bp) deletion and an in-frame 6 bp insertion in *PK-LR* gene have been described in PK-deficient Basenjis[5] and WHWTs,[10] respectively. Here, we report on the mutation found in the previously described PK-deficient Cairn Terrier[12] and 3 additional mutations causing PK deficiency, each in a different breed of dogs, and the associated clinicopathologic features of the disease, including severe hepatic hemochromatosis. Screening for these newly identified mutations will simplify the diagnosis of PK deficiency in those breeds and allow carrier detection.

Methods

Animals and Samples

EDTA blood samples from dogs with chronic unexplained regenerative anemia, a negative direct Coombs' test, as well as related dogs and other dogs from the same breed were studied, when available. Labrador Retrievers (2 siblings, 5 unrelated); Pugs (2 siblings, 1 unrelated); and Beagles (n = 68, 39 anemic, 10 genetic screening, 19 unknown reason), WHWTs (n = 248, 22 anemic and 226 non-anemic) and Cairn Terrier (n = 1) were studied. Anemic Labrador, Pug, Beagles, WHWT and Cairn Terrier samples were referred to PennGen Laboratory (http://research.vet.upenn.edu/penngen) from primary veterinarians and some WHWT and Beagle samples were mailed in from breeders from the USA for general testing and breeding purposes. With an exception of a few WHWT samples that were mailed in from Brazil and the UK. The WHWT and Beagle samples were collected between 2000 and 2011. In addition, EDTA blood and cheek swab samples, sent to PennGen for screening of known PK mutations, were included. These dogs were frequently first treated for immune-mediated hemolytic anemia (IMHA) and infectious diseases, but had a negative direct Coombs' test result when tested and did not respond to treatment. The studies were approved according to the guidelines of the Institutional Animal Care and Use Committee at the University of Pennsylvania.

Clinicopathologic Studies

Routine clinical examination, laboratory test results, and imaging information were reviewed, when available. Erythrocytic PK activity,[3,6–9] osmotic fragility (OF),[14] and hemoglobin-oxygen ($Hb\text{-}O_2$) dissociation curves[6,30] were determined in some dogs and compared with simultaneously tested samples from healthy control dogs of different breeds as previously described. Erythrocytic 2,3- diphosphoglycerate levels and PK isoforms could not be measured because of the current unavailability of kits and reagents. A

routine necropsy was performed on 1 PK-deficient Labrador Retriever, with histopathologic evaluation of relevant formalin-fixed, paraffin embedded tissues using routine hematoxylin and eosin (H&E) staining. Hepatic lesions were additionally evaluated with Perl's iron stain and iron quantitation in parts per million dry weight as determined by inductively coupled-mass spectrometry.

Genetic and PK Sequencing Studies

Genomic DNA and RNA extraction

Genomic DNA (gDNA) was isolated from 200 μL EDTA-anticoagulated blood and cheek swabs using Generation Capture Column kit (Cat. #: 159916)[a] and Gentra Puregene Buccal Cell kit (Cat. #: 158867),[a] respectively, according to standard genetic methodologies. Genomic DNA was also extracted from formalin-fixed paraffin-embedded (FFPE) kidney and spleen samples with a QIAamp DNA FFPE Tissue kit (Cat. #: 56404).[a] A column-based QIAamp RNA Blood Mini kit (Cat. #: 52304)[a] was used to extract RNA from fresh EDTA blood, and an Ambion RiboPure^{TM}-Blood kit (Cat. #: AM1928)[b] was used to extract RNA from EDTA blood stored in RNAlater. Fresh RNA was reverse-transcribed into complementary DNA (cDNA) using a High Capacity cDNA Reverse Transcription kit (Part #: 4368814).[c] Genetic nomenclatures used in this article are in accordance with the recently updated version of the Guidelines and Recommendations for Mutation Nomenclature by the Human Genome Variation Society.[31]

Amplification of PK-LR gene and R-PK cDNA

Genomic DNA covering all exons of the *PK-LR* gene and the entire R-PK cDNA were amplified by using the *Taq* DNA polymerase (Cat. #: 18038- 042)[d] and Takara *Taq* Hot Start (Cat. #: R007A)[e] by polymerase chain reactions (PCR) with primer pairs complementary to gDNA and cDNA using annealing temperatures as listed in Table 1. ThermalAce DNA polymerase (Cat. #: E0200)[d] was utilized to overcome the difficulties in amplifying the GC-rich exons 4–6.

Table 1.

Genomic DNA and cDNA primer pairs, product lengths, and annealing temperatures used for the amplification of *PK-LR* gene from anemic dogs.

	Exons	Forward (5′–3′) Primer	Reverse (5′–3′) Primer	Annealing Temperatur (°C)
gDNA Primer Pairs	Exon 1	GGCAGAGTGCCAGCAAGTCTCA	GCCAGCAGCAGATGTTGGGCAT	63
	Exon 3	TGCTGAGGTGAAGGGTCGC	AGGGGGAAAACATCTTGGGAAACT	54
	Exons 4–5	TGGCTGGACTATTGTGGCGTGG	TAGCTCCACTTCGGTCTCAGGG	70
	Exon 6	TGGACACCAAGGGACCGGAGAT	ACAGGCGTAGGAGAGTGTGTGG	70
	Exons 7–8	GGGGGCTGGCGGGTTATCT	TCATCCGAAAATAGGGCTCACA	63
	Exons 9–10	GCTTGGGCAAGTCCTCAGTC	CCTCTTGCCCCAGAATGACAG	67
	Exon 11	GAGGCATTGCAGAAGGGC	GAGAAAATGTGGGATCAGAGGA	62
	Exon 12	CCCCCACCAGCTGGAACTAA	GTGACTTTGTGGAGATGCCCTG	62
cDNA Primer Pairs	A (Exons 1–5)	CCCCAGGAGCTTTGGTCAAGGA	GTTGCAAAGCTCTCCACGGCCT	58
	B (Exons 4–7)	ATTGCGCGCCTCAACTTCTCCC	AGGTTTACGCCTTTCCGGCTGC	60
	C (Exons 6–10)	TAGTGAAGGGCTCCTGGGTCCT	TTCCTCAAACAGCTGCCGGTGG	59
	D (Exons 9–12)	CCGCCAAGGGCAAATTTCCTGT	GGATATGCTGAGCACTCGCATGA	62

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Each reaction was prepared to a final volume of 50 μL containing 2 μL of gDNA, 5 μL of 10× PCR buffer, 2 μL of dNTPs mixture (10 mM), 2 μL $MgCl_2$ (50 mM), 0.5 μL primer pairs at 10 mM concentration, and 0.25 μL of *Taq* DNA polymerase (5 U/μL), with the exception of samples from Beagles, which were amplified with Takara Ex *Taq* Hot Start (Cat. #: R006A).[e] The exons 4, 5, and 6 were amplified at the same final volume and gDNA concentrations, but with 5 μL 10× ThermalAce buffer, 1 μL of 50× dNTP mix (10 mM), 0.5 μL primer pairs at 10 mM concentrations, and 1 μL ThermalAce DNA *Taq* (2 U/μL). Amplification of R-PK cDNA was performed with a slightly different master mix using 0.25 μL of Takara *Taq* HotStart (5 U/μL) and 4 μL of dNTPs mixture (2.5 mM).

Amplifications were performed with the following sequential steps for the gDNA: denaturation at 95°C for 5 minutes, followed by 30 cycles of amplification with initial denaturation at 95°C for 30 seconds, annealing of primers at corresponding temperature for 30 seconds and extension at 72°C for 30 seconds, and a final extension at 72°C for 7 minutes. Cycling conditions for cDNA samples were the same as for gDNA with the addition of a 1°C touch-down for 10 cycles to the corresponding annealing temperature of each primer pair.

Sequencing of amplified DNA products

Amplified products were analyzed using 6% polyacrylamide gel, 1.5% agarose gel electrophoresis, or both. Products were purified with a columnbased QIAquick PCR Purification kit (Cat.#: 28104)[a] or a QIAquick Gel Extraction kit (Cat. #: 28704).[a] Purified PCR products were sequenced at the University of Pennsylvania School of Medicine's Core DNA Sequencing Facility using standard amplification primers. The DNA sequences were assembled and compared with *PK-LR* gDNA (Gene ID: 490425)[5,10] and R-type cDNA (NM_001256262.1)[32] and also to the previously published Basenji and WHWT sequences using the DNASTAR[f] and ABI Sequence Scanner software, UCSC Genome Browser[33] and NCBI-Blast[34] web-based programs. Furthermore, the R-PK protein was analyzed by SIFT[35] to determine if the mutations were tolerated and by Perspired[36] protein structure prediction server to locate their positions with respect to previously published active sites.[37]

Breed specific mutation assays

To screen for the discovered RPK mutations in Labradors and Pugs, a specific primer pair (Forward: 5′-GGCTGGAGACCCAAGTGGAGAA-3′; Reverse: 5′-GATGGCAGCCACATCACTGGCT-3′) was used to amplify a region in exon 7 that produces a 188 bp product from either gDNA or cDNA. This PCR product was subsequently

digested for 2 hours with restriction enzyme *MwoI* (Cat. #: R0573L)[g] at 60°C for Labradors or *PflFI* (Cat. #: R0595L)[g] at 37°C for Pugs, and then analyzed by 6% polyacrylamide gel electrophoresis.

The mutation causing R-PK deficiency in Beagles did not generate or eliminate a known restriction enzyme recognition sequence. Therefore, a primer pair (Forward: 5′-GAAATCCTGGAGGTGAGTGcC- 3′ and Reverse: 5′-TGCACCGACCAATCATCATCTTCT- 3′) that introduces a single base mismatch (lowercase c instead of an A at the second to last nucleotide in the forward primer) near the mutation site was used to create a recognition site for *NgoMIV* (Cat. #: R0564L)[g] for the normal allele. The PCR product was digested with *NgoMIV* at 37°C for at least 2 hours and then analyzed by 6% polyacrylamide gel.

In PK-deficient WHWTs, a 6 bp insertion in *PK-LR* exon 10 has previously been identified.[10] Using the previously described methods and primer pair surrounding the insertion (Forward: 5′-CCCACGGAGGTCACTGCCATA-3′ and Reverse: 5′-GCTCTGTTGGTGTTGCCAGTG- 3′), the difference between alleles is readily differentiated by size on a 6% polyacrylamide gel. Splenic tissue from the previously published PK-deficient Cairn Terrier[12] was also tested for the PK mutation in WHWTs.

Results

Clinicopathologic Results

A 2-year old male rescued Labrador Retriever (Lab 1), an 8-month-old female Pug (Pug 1), and a 10-month-old female Beagle (Bgl 1), all of which presented with chronic severe anemia with occasional hemolytic crises and osteosclerosis, were further examined to determine their molecular genetic defects. Despite severe anemia and pallor, the signs were generally mild except for exercise intolerance and occasional episodes of severe weakness. The anemia was highly regenerative, macrocytic, and normo- to hypochromic (Table 2). Blood smears revealed marked polychromasia and anisocytosis with few echinocytes. There was no evidence of blood loss. There was hyperbilirubinuria and mild hyperbilirubinemia but no icterus. Routine testing did not reveal any acquired causes of hemolytic anemia: drug and chemical exposure was excluded historically; serology tests were negative for infections known to cause hemolysis; and the direct Coombs' test results were also negative. Furthermore, the dogs failed to respond to administration of doxycycline and prednisolone and other immunosuppressive treatment. The PK-deficient WHWT[9–11] and the Cairn Terrier[12] have similar clinicopathologic signs to the cases as previously described.

Table 2.

Hematologic parameters in dogs with PK deficiency.

	Labrador (7)[*]	Pug (2)	Beagle (2)	WHWT (7)	Reference Interval[a]
HCT (%)	10–13	24–26	13–29	17–28	40–60
HGB (g/dL)	2–4	7–8	4–8	5–9	13–21
MCV (fL)	88–94	89–91	77–86	75–92	63–76
MCHC (g/dL)	28–32	29	28	28–35	32–36
Reticulocyte Count ($\times 10^3$/μL)	102–254	681[**]	415–771	656[**]	<60

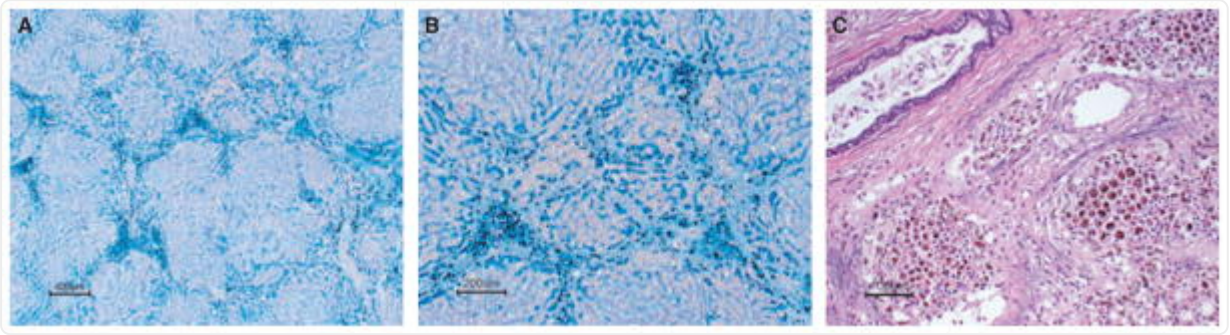
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[a]Reference interval from the clinical pathology laboratory at the Veterinary School of University of Pennsylvania.
[*]Number of times assayed for each breed.
[**]Available only on 1 occasion before diagnosis, but persistent severe polychromasia observed.

The erythrocytic OF curves from all studied dogs were of normal sigmoid shape; the point of 50% hemolysis was only slightly increased compared with normal, suggesting a mildly increased OF. The Hb-O_2 dissociation curves were shifted to the right, resulting in a higher P50 compared with the simultaneously run control, indicating a reduced oxygen affinity. Because of the anomalous expression of M-type PK in erythrocytes from PK-deficient dogs, erythrocytic PK activities in affected animals were above normal (eg, Lab 1– 215% compared with the 100% control). Iron parameters, measured in serum from Lab 1, revealed severe hyperferritinemia (2,661 ng/mL, reference interval 80– 800 ng/mL) and high serum iron concentration (355 μg/dL, reference interval 88–238 μg/dL) with a near complete iron saturation (total iron binding capacity 366 μg/dL [reference interval 245–450 μg/dL] and 97% saturation [reference interval 20–60%]). Radiographs of either the femur or humerus showed increased cortical bone and/or increased blotchy bone density in the marrow areas. Abdominal radiographs and/or ultrasound indicated mild to moderate hepatosplenomegaly.

Labrador 2 (the littermate of Lab 1 living in the same household) presented with similar clinicopathologic findings, including severe regenerative anemia and osteosclerosis of the long bones. He was euthanized at 2 years of age because of progressive clinical signs and poor prognosis, and a complete diagnostic necropsy was performed. Relevant gross findings included diffuse icterus, dense cortical and trabecular bone, scant bone marrow which sank in water, and hepatosplenomegaly. The liver in the dog weighed 1,070 g (6.7% of body weight, normal canine reference interval 3.2–3.7%), with a dark red-brown granular capsular surface and multifocal circular firm tan foci 1 –2 mm in diameter within the parenchyma. Severe iron accumulation was observed in form of hemosiderosis and hemochromatosis, with a postmortem liver iron of 37,300 ppm analyzed on a dry weight basis (reference interval: 350–1,200 ppm dry weight), and congruent histologic findings. These included bridging portal fibrosis with lobular atrophy and regenerative nodules, hyperplastic bile duct profiles and increased numbers of macrophages and Kupffer cells containing abundant golden brown pigment (hemosiderin) on routine H&E. Perl's iron staining confirmed the presence of iron within Kupffer cells and to a lesser extent, in hepatocytes (Fig 1A and B). Hemosiderosis and erythrophagocytosis were also present within medullary cords of hepatic lymph nodes (Fig 1C). Bone marrow analysis showed a decreased myeloid to erythroid (approximately 1 : 4) ratio with erythroid hyperplasia, mild fibroplasia and hemosiderosis. The cortical bone was mildly thickened.

Fig 1.



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Histopathology of liver and hepatic lymph node from PK-deficient Labrador (Lab 2). (A) Perl's iron stain of hepatic parenchyma at 20× and (B) at 40×, shows accumulation of dark blue granules of hemosiderin within Kupffer cells and blue tint of iron within hepatocytes. (C) Hematoxylin and eosin of hepatic lymph node (100×), with medullary cords expanded by large numbers of macrophages that contain coarse, dark brown refractile granules (hemosiderin) and intracytoplasmic fragments of erythrocytes (erythrophagocytosis). Scattered foci of extramedullary hematopoiesis are also present.

Additional blood or cheek swab samples submitted to the laboratory were available for study: 1 female littermate of Pug 1 (Pug 2) which was not anemic as well as other pugs (Pug 3 and 4); 5 Labrador Retrievers, one of which had a Coombs' positive anemia and others with less well defined but poorly regenerative anemias; 39 anemic and 29 other Beagles (tested for genetic screening [n = 10], unknown reason [n = 19]); and 22 anemic and 226 non-anemic WHWTs. Many of the anemic dogs were suspected to have PK deficiency, whereas the others were examined because of the breed or the relatedness to affected dogs. No specific hematologic information was available for Pug 1 and sibling and Bgl 1 probands.

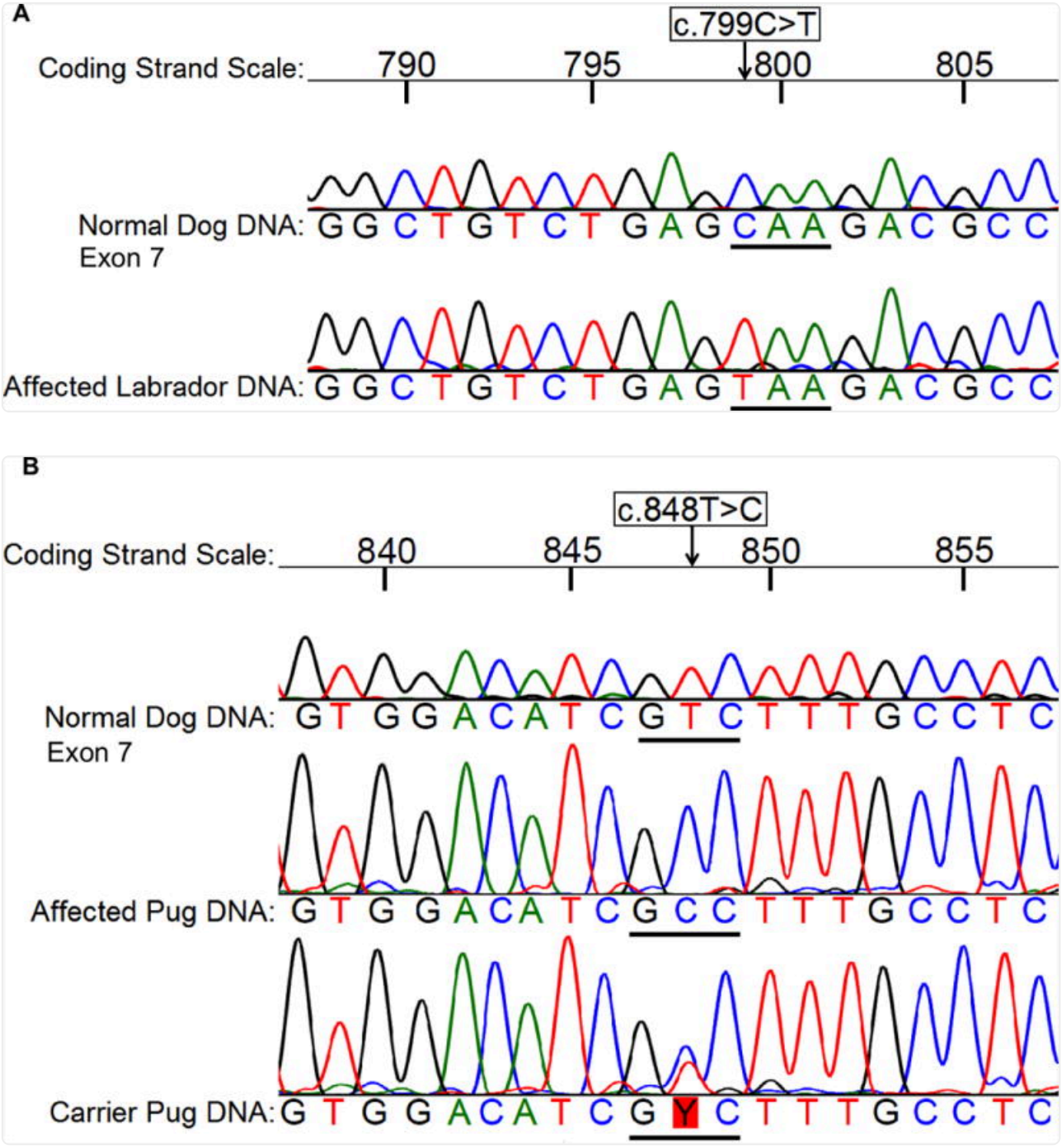
Sequencing Results and Mutation Analysis

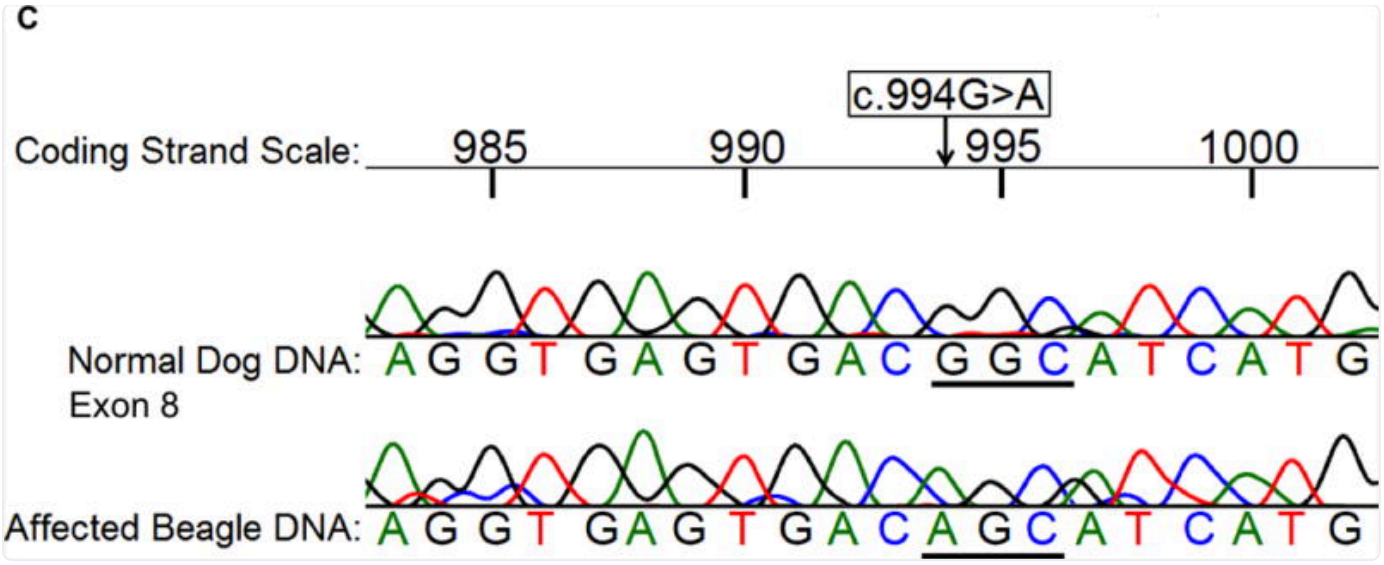
Except for the specific mutations described below, the exonic DNA and thus the protein sequences from the PK-deficient Beagle, Labrador, and Pug were completely identical to that of the canine genome sequence and the R-type cDNA (NM_001256262.1) and also to the previously published normal WHWT sequence[10] but differed by 4 bp to the partial coding sequence originally described in normal Basenjis.[38]

Labrador Retriever

Sequencing of Lab 1 and 2 revealed a homozygous single base substitution in exon 7 (c.799C>T) of the *PK-LR* gene (Figs 2A and 3), which changes a glutamine (CAA, Gln) into a stop codon (TAA, *), leading to a severe truncation of the R-PK protein (p.Gln267*) compared with the full-length normal canine R-PK protein of 574 amino acids. Because of the early termination, the enzyme from these 2 Labrador siblings lack most of the active sites. Five other Labradors with anemia, one of which had Coombs' positive (4+ agglutination) IMHA, were also sequenced for the mutation, but they were homozygous for the wild-type allele.

Fig 2.

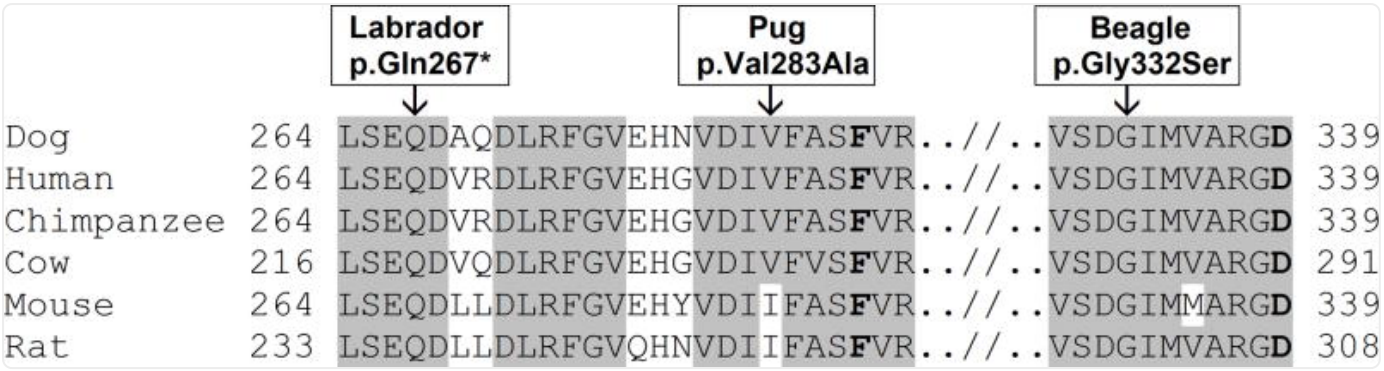

A
c.799C>T
Coding Strand Scale:
790
795
800
805
Normal Dog DNA:
G G C T G T C T G A G C A A G A C G C C
Exon 7
Affected Labrador DNA:
G G C T G T C T G A G T A A G A C G C C
B
c.848T>C
Coding Strand Scale:
840
845
850
855
Normal Dog DNA:
G T G G A C A T C G T C T T T G C C T C
Exon 7
Affected Pug DNA:
G T G G A C A T C G C C T T T G C C T C
Carrier Pug DNA:
G T G G A C A T C G Y C T T T G C C T C



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Sequence tracings of gDNA regions in exons 7 and 8 of the canine *PK-LR* gene. The wild-type sequence is compared with sequences from a PK-deficient Labrador (A), Pug (B), and Beagle (C). Mutation positions are indicated with an arrow on the coding strand scale and the codon with the mutation is underlined for each breed.

Fig 3.



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Amino acid sequence homology among mammals and the sites of canine R-PK mutations. The analysis shows that the R-PK amino acid sequences are highly conserved among dog, human, chimpanzee, cow, mouse, and rat at 3 mutation positions (267-Labrador, 283-Pug, and 332-Beagle). The conserved areas around the mutations are shaded, and the active sites are shown in bold capital letters. The polymorphism [isoleucine (I)] at position 283 for the mouse and rat, is a tolerated amino acid change by the SIFT protein prediction analysis.

Pug

In Pug 1, a different homozygous single base substitution was found in exon 7 (c.848T>C) of the *PK-LR* gene sequence (Figs 2B and 3). This point mutation changes GTC, which codes for valine (Val), into GCC, coding for alanine (Ala). This p.Val283Ala missense mutation is not tolerated based on SIFT analysis and resides in a β-strand area of *PK-LR*, a few amino acids away from the substrate (phospho-enolpyruvate) binding site in domain A. Based upon sequence comparison of exon 7, this residue resides in a highly conserved region among mammalian species. Indeed, only Val and isoleucine (Iso) are tolerated at position 283; higher mammals have a Val, whereas rodents have an Iso at this position (Fig 3).

Beagle

A single base substitution mutation was discovered in exon 8 (c.994G>A) of the *PK-LR* gene in Bgl 1. This point mutation changes the codon GGC to an AGC and thus replaces a glycine (Gly) with a serine (Ser) (Figs 2C and 3). According to SIFT analysis, only Gly is tolerated at this position. As all mammals have a Gly at this position, it is highly likely that the p.Gly332Ser missense mutation causes a nonfunctional R-PK protein. Moreover, the sequence around the site of this missense mutation is also highly conserved

throughout all species analyzed and is only a few amino acids away from the Mg^{2+} binding site in domain A (Fig 3).

Cairn Terrier

This previously described PK-deficient dog with severe anemia, osteosclerosis, and hemochromatosis[12] was found to have the same 6 bp insertion as reported for the PK-deficient WHWT dogs (data not shown).

Other Breeds

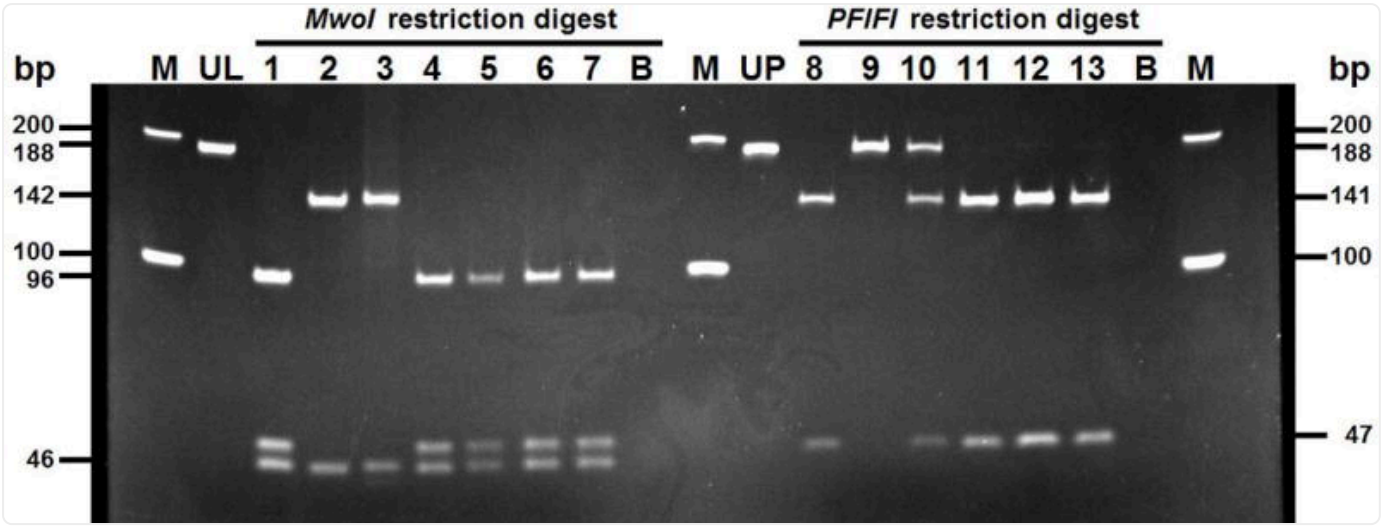
The complete R-PK coding sequences from a Spitz and Chihuahua suspected to have anemia because of PK deficiency did not reveal any mutations in their gDNA (data not shown).

Breed-specific Mutation Test Results

Labrador PK test

Digesting the exon 7 PCR product of 188 bp with the restriction enzyme *MwoI* will cause the wild-type allele to be cut at 2 recognition sites with 3′ overhangs of 3 bases and will produce 3 fragments of 46, 46, and 96 bp on the coding strand. The mutation observed in the Labrador Retrievers obliterates the second recognition site, resulting in 1 cut that produces only two fragments of 46 and 142 bp. Thus, only 2 bands will be observed for the mutant allele, allowing for easy differentiation of the wild type and mutant alleles by the presence of a 96 bp band versus 142 bp band. Lab 1 and 2 only displayed the 142 bp fragment, whereas all the other dogs analyzed, which included 1 healthy Labrador, 3 PK-deficient/ healthy Pugs, and 1 healthy mixed-breed dog, showed the wild-type 96 bp fragment (Fig 4).

Fig 4.



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Labrador and Pug mutations analyzed by restriction enzyme digestion followed by electrophoresis on polyacrylamide gel (6%). Lanes: M: 100 bp DNA marker: UL and UP: uncut fragment from affected Labrador (Lab 1) and Pug (Pug 1), respectively; 1 and 12: healthy control Labrador; 2 and 13: affected Lab 1 (proband); 3: affected Lab 2; 4 and 11: healthy control of different breed; 5 and 8: healthy control Pug; 6 and 9: affected Pug 1 (proband); 7 and 10: carrier Pug; B: blank. Lanes UL and UP at 188 bp are the PCR amplification products of exon 7 from affected Labrador and Pug dogs before digestion. Products are digested with *MwoI* for Labradors, which cuts the wild-type allele twice producing 3 fragments of 96, 46, and 46 bp (lanes 1). Because of the 3 base 3′ overhangs, the center 46 bp fragment has 3 bp overhangs on both sides thus migrates at slower pace and appears slightly larger. The affected Lab 1 and Lab 2 are cut only once by *MwoI* resulting in 2 DNA fragments of sizes 142 and 46 bp (lanes 2, 3). The digestion of the PCR products with *PFlFI* for the Pug produces two bands of 141 and 47 bp for the normal allele (lane 8), however, the mutant allele is missing the restriction site and does not get cleaved, thus showing the 188 bp uncut DNA band (lane 9). The carrier Pug shows both mutant and wild-type alleles and thus has 3 bands (lane 10). Amplification products from different breeds of dogs (lanes 4,11) show only the wild-type allele digestion pattern. Digesting the PCR products from Labradors with *PFlFI* and vice-versa for the Pug validates the specificity of the restriction enzyme digests: Pug samples (lanes 5, 6, 7) for Labrador digest and Labrador samples (lanes 12, 13) for Pug digest.

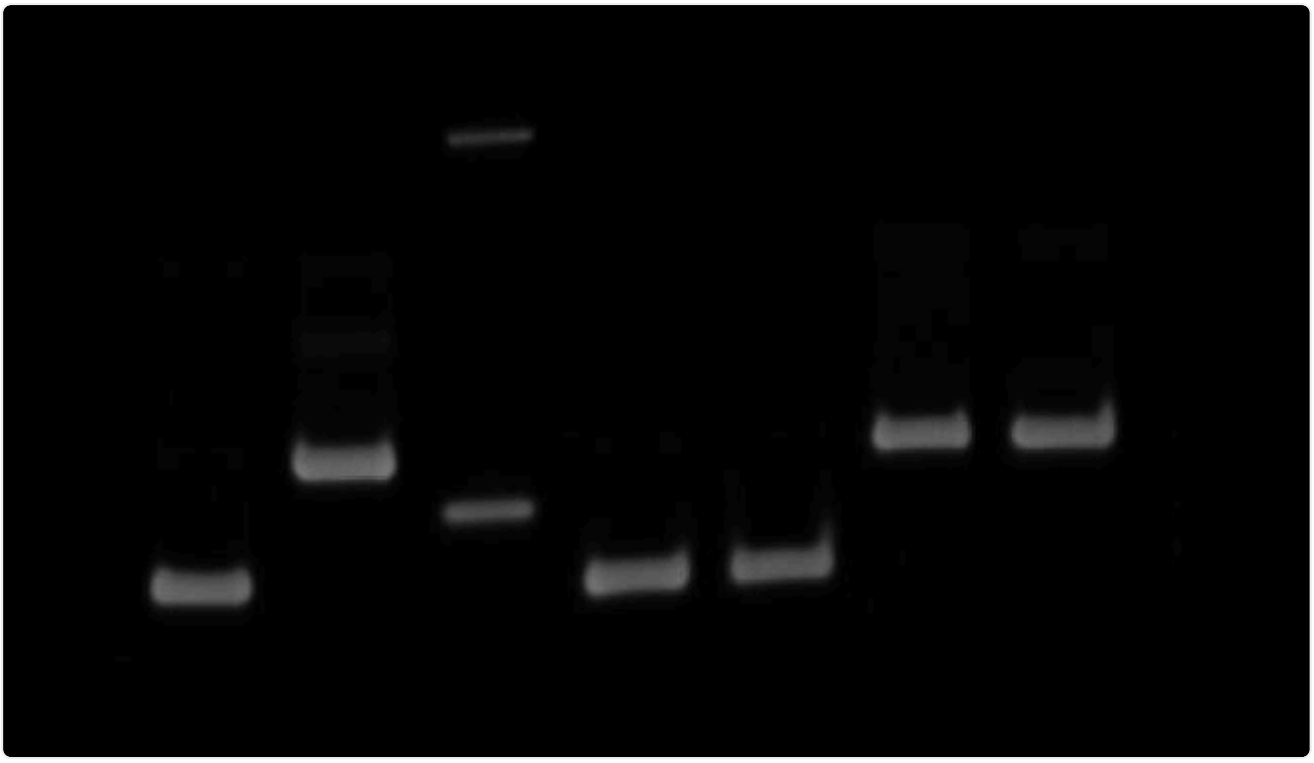
Pug PK test

When the 188 bp PCR product used for analysis of the mutation in PK-deficient Labradors (described above) is digested with the *PflFI* restriction enzyme, the normal allele is cut into 141 and 47 bp fragments. In contrast, the mutant allele from the PK-deficient Pug 1 is not digested. The sample from Pug 2 displays all 3 bands of 188, 141, and 47 bp on the polyacrylamide gel, and thus has both alleles (carrier). In contrast, the other 2 Pugs (Pug 3 and 4) and PKdeficient Labradors tested showed a normal digestion pattern (Fig 4).

Beagle PK test and screening

The digestion of the 109 bp PCR product of exon 8 with the *NgoMIV* enzyme cuts the wild-type allele into 19 and 90 bp fragments, whereas the mutation in Beagles abolishes this restriction site. Carrier animals exhibit all 3 fragments (19, 90, and 109 bp) and thus reveal both alleles on a 6% polyacrylamide gel (Fig 5). A biased group of samples were submitted, primarily from Beagles suspected of PK deficiency. Samples from Bgl 1 and 38 other anemic Beagles from the United States were homozygous for the mutant allele. The average age at the time of diagnosis for the PK-deficient Beagles was 2 years, and ranged from 7 months to 9 years (also represents the oldest PK-deficient Beagle). Among the 29 additional Beagles screened, 2 Beagles were classified as carriers and the rest as clear. The mutant allele frequency in this biased group of samples was estimated to be 0.37 by the Hardy-Weinberg equation.[39]

Fig 5.



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Amplification products from healthy and PK-deficient Beagles analyzed by restriction enzyme digestion followed by electrophoresis on polyacrylamide gel (6%). Lanes: M: 100 bp DNA marker; 1 and 2: Normal and affected controls, respectively; 3 and 4: anemic Beagles in duplicates: N: Normal: A: Affected. The restriction digest with *NgoMIV* cuts the 109 bp wild-type allele into 90 and 19 bp fragments; however, the mutant allele is not digested thus only shows the uncut band at 109 bp. The 19 bp fragment is indistinguishable from primers and therefore is not shown in this figure.

WHWT PK test and screening

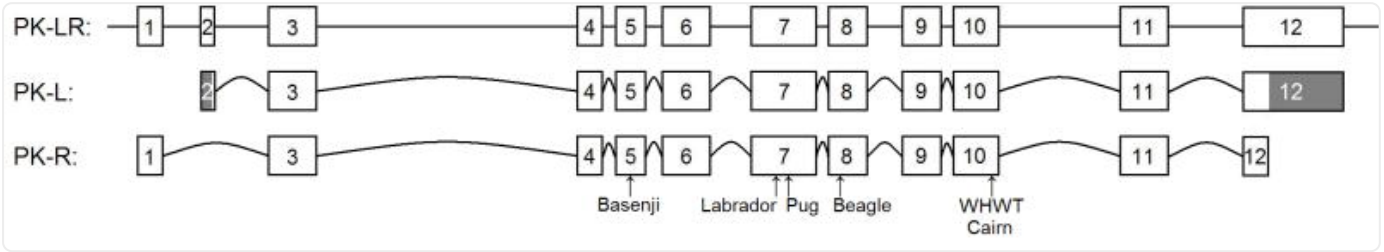
As a result of the 6 bp insertion in exon 10, which causes an in-frame addition of 2 amino acids (threonine and lysine), the mutant allele in WHWTs can be readily differentiated by the larger 123 bp product compared with the normal 117 bp fragment in other dogs on a 6% polyacrylamide gel.[10] Among the 248 WHWTs tested, 9% were found to be homozygous for the mutant allele and were anemic and 35% showed both alleles (carriers). The mutant allele frequency in this biased group of samples was estimated at 0.26.[39] The average age at the time of diagnosis of PK deficiency in WHWTs was 1.5 years, ranging from 2

months to 5 years of age. Most affected WHWTs were found in the United States but also in Brazil,[11] England and Ireland[40]; the oldest PK-deficient WHWT died at 9 years of age.

Discussion

Typically, only 1 disease-causing mutation is observed in an inborn error of metabolism affecting a single breed of dog. However, more recently, different breed-specific mutations have been identified in a single gene, such as von Willebrand disease.[41] Alternatively, a single ancestral mutation can be responsible for a specific defect in different, but often related breeds of dogs, such as the multi-drug resistance 1 gene mutation[42] or factor VII deficiency.[43,44] After the early discovery of different mutations in *PK-LR* in the Basenjis[5] and WHWTs,[10] we describe in this report 3 additional breed-specific *PK-LR* mutations in the Labrador Retriever, Pug, and Beagle, making this gene the most frequently mutated canine gene observed to date (Fig 6). Not surprisingly, the Cairn Terrier, which is a closely related breed to the WHWT, with PK deficiency had the same insertion mutation as the previously, and herein, described PK-deficient WHWTs.

Fig 6.



Open in a new tab

Structure of the canine *PK-LR* gene, transcripts of liver (NM_001256018.1) and erythrocyte (NM_001256262.1) splicing variants and the known breed specific PK mutation sites. Although the *PK-LR* gene has 12 exons, the different tissue-specific splicings reduce the exon count to 11 for both L-PK and R-PK. Exons and introns are drawn to approximate scale. Exons are numbered and depicted by rectangles. The 5′ and 3′ gray regions in the L-PK transcript are untranslated regions of that splice variant as per NCBI. Arrows indicate the locations of 5 mutations associated with PK deficiency, including the single bp deletion in exon 5 in the Basenji, and the 6 bp insertion in exon 10 in the WHWT and Cairn terrier, and the newly identified nonsense mutation in the Labrador (c.799C>T) and missense mutations in Pug (c.848T>C) and Beagle (c.994G>A).

In humans, PK deficiency is also the most common erythroenzymopathy and at least 180 mutations have been described.[25] Although most mutations are observed in the second half of the *PK-LR* gene,[45] mutations are spread throughout the entire gene with no particular hot spot areas. They include single base changes resulting in amino acid substitutions, insertions, and deletions with or without early termination of the RNA.[19,20,23–27] The location and type of PK mutation can predict the severity of the disease presentation in a particular PK-deficient human patient. As expected, non-sense mutations and large deletions causing early truncation of the predicted protein lead to more severe phenotypes with greater transfusion dependence than certain missense mutations.[27,45]

Similarly, PK mutations in dogs include a single base deletion (Basenji),[5] a 6 bp insertion (WHWT[10] and Cairn Terrier reported here), a nonsense single base substitution (Labrador) and 2 missense mutations (Beagle and Pug). Although site-specific mutagenesis or expression experiments have not yet been performed for any PK mutation in dogs (and are seldom done for human mutations), the evidence to date strongly suggests that these specific mutations will have a negative impact on the protein length and stability, as well as enzyme function. Most notably, instead of producing a normal canine R-PK cDNA and protein sequence of 1725 bp and 574 amino acids, respectively, both the deletion[5] and nonsense substitution reported here result in early stop codons and severe truncation of the protein. The predicted protein fragments (166 and 266 amino acids) are less than half the normal enzyme size and therefore are likely unstable and undoubtedly dysfunctional. In contrast, the 6 bp in-frame insertion located at the C-terminal end in WHWTs and Cairn Terriers results in the addition of 2 amino acids that alter the activation site,[10] suggesting that it will impact function, but may not affect stability. Similarly, the here observed missense mutations (Beagle and Pug), situated in the middle of the protein, appear to impair function, as they are not tolerated in modeling and SIFT analysis, but are unlikely to affect protein stability. Thus, the insertion and 2 missense mutations would be predicted to cause less severe PK-dysfunction.

Based upon the clinical assessment and anecdotally reported longevity of PK-deficient dogs, it appears that Basenjis and Labrador Retrievers have indeed a more severe hemolytic anemia leading to earlier death than do other breeds. In particular, PK-deficient WHWTs[9,11] and Beagles[7,8] have been observed to live 9 years despite being persistently severely anemic, whereas Basenjis[3,6] and the PK-deficient Labrador Retrievers reported here seem to be more severely affected and do not exceed 5 years of age, even with appropriate supportive treatment. At the time of diagnosis, the Pug was only 11 months old and at the time of this report she is 2 years of age and is compensating well despite severe anemia.

Chronic severe hemolytic anemia in a breed known to have PK-deficiency or in any persistently anemic dog with massive reticulocytosis, a negative direct Coombs' test, and lacking known infectious diseases should include PK deficiency in a differential diagnosis. [2–12] Notably, PK deficiency in anemic dogs is often accompanied by a unique, yet unexplained, development of osteosclerosis, which was observed in the PKdeficient dogs studied here. Osteosclerosis develops at about 1 year of age and is progressive in PK-

deficient dogs[3–12] but is never seen in humans[27] and cats[14–18] with PK deficiency, nor with any other hemolytic anemia in dogs and other species.

However, knowledge of the breed-specific mutations is the most important tool for screening a purebred dog and clearly simplifies the diagnosis of PK deficiency. For each of the known mutations, the DNA around the mutation can be amplified and the fragment either sequenced or otherwise analyzed for the specific mutation. Although insertions and deletions can be directly identified by size on a gel, missense mutations often require a restriction enzyme analysis. In fact, for the Beagle mutation, a restriction site had to be created by introducing an additional base change in the forward primer very close to the mutation. Notably, these simple DNA based tests are far more accurate than the cumbersome enzymatic assays, which are complicated by the presence of M-type PK and are not routinely performed by any veterinary laboratory. [3,4] They also only require a small amount of sample (EDTA blood or cheek swab) which can be simply sent by regular mail.

Although functional characterization of these mutant enzymes would provide mechanistic insight into the disease, there are unfortunately no readily available means to assess R-PK activity in vitro, as all PK-deficient dogs have an anomalous M_2-PK expression in erythrocytes, which is nonfunctional in vivo, but obscures erythrocytic enzyme activity measurements in vitro.[2–12] Currently there is also no specific antibody available that recognizes canine R-PK and M^2-PK. Although PK activity measurement after electrophoresis and heat exposure have been used for indirect differentiation of the 2 isoforms in erythrocytes,[2,3,8] the PK activity in affected dogs was not further evaluated beyond the total erythrocytic enzyme activity reported here.

In humans, among the 180 *PK-LR* mutations published, a few occur more frequently in specific ethnic groups.[22] For example, the c.1436G>A is most frequently observed in both White and Japanese populations, the c.1456C>T in populations from Italy, Spain, and Portugal, the c.1468C>T in Asian populations, and c.1529G>A in US and Northern/Central European populations.[22,27] The c.1436G>A is also wide-spread in the Amish community in Pennsylvania[19] and Ohio,[20] who are routinely screened for this mutation. On the other hand, no formal surveys have assessed the prevalence of *PK-LR* mutations in dogs. The originally reported mutation in Basenjis[3–6] appears to have been eliminated from the breed as no affected animals are apparently seen clinically and in the screening laboratories. The PK mutations in Labrador Retrievers and Pugs seem to be limited to a few dogs and possibly a single family, whereas the other PK mutations seem to be common and even wide-spread within a breed. In particular, PK-deficient WHWTs[9,10] and Beagles have been found throughout the United States and affected WHWTs were also identified in Europe[40] and South America.[11] However, it should be noted, that the frequency of the different PK mutations reported here is highly biased by the selected samples received. After an index case has been identified, it depends on the owner and breeder, who are informed about the results, to alert others of the possibility that a chronically anemic related dog may be affected and others may be carriers. Moreover, some breeders and breed clubs may embrace the opportunity to screen their breeding animals for a

known PK mutation, whereas others will ignore this entirely. The frequency assessment in dogs is further hampered by the lack of a mandatory open registry of tested dogs. In this report, the dogs screened were highly biased toward anemic and related dogs. Finally, it is likely that additional mutations will be found in other breeds with suspected PK-deficiency, such as the Chihuahua and Spitz. Although we have not as yet identified a mutation in the coding sequence of these dogs, promoter and splicing defects have not been ruled out. Thus, additional PK mutations are likely to be found. Similarly, after the original description of a nonsense mutation in the *PFKM* gene causing PFK deficiency in English Springer[46] and Cocker Spaniels[47] with hemolytic crises, the same mutation was recently also seen in PFK-deficient Whippets,[48] but a different missense mutation was recently discovered causing PFK deficiency in Wachtelhunds.[49]

Increased deposition of iron in the parenchyma of organs without clinical manifestation is termed hemosiderosis, and hemochromatosis when hepatic dysfunction occurs.[50,51] Primary or hereditary hemochromatosis is caused by several different mutations in the hemochromatosis (*HFE*) gene in humans[52] but has not been documented in dogs. A rare combination of inherited PK deficiency and specific *HFE* gene mutations has been described in a few human patients.[28,53] Genetic defects in the genes associated with hemochromatosis were not studied in these Labradors. However, humans and other animals with chronic hemolytic anemia are anyway at risk for developing iron overload particularly when chronically transfused over many years. This secondary hemosiderosis and hemochromatosis has been associated with PK deficiency in dogs,[12] cats,[14,17] and humans.[3,27,28,50,54–56] In healthy Basenjis, the apparent half-life of erythrocytes as measured by radiolabelled ^{51}Cr is approximately 20 days, but only 2.5–3 days in PK affected dogs.[57] Reduced erythrocyte survival results in a plasma iron turnover rate that is estimated to be twice normal.[57] In the PK-deficient Labrador Retrievers, extremely high serum iron values and hepatic iron were found and hemosiderosis and hemochromatosis were documented histologically in the one dog necropsied. The postmortem liver iron level in the necropsied dog was 31× the normal upper reference interval, resulting in the changes observed within the hepatic parenchyma. Although the Labrador Retrievers studied here received only occasional blood transfusions, they were fed a meat-based diet, which may have augmented the amount of iron absorbed. Clearly, any iron supplementation is contraindicated in these PK-deficient patients as well as in most other patients with chronic hemolysis (except when large amounts are lost in the urine because of intravascular hemolysis). Indeed, fibrosis and cirrhosis develop in people at an iron threshold level of approximately 22,000 ppm per dry weight matter, which is 12 × normal.[58] To reduce such detrimental effects, human patients with severe hemolytic anemia because of R-PK deficiency routinely need iron chelation treatment. However, as hemochromatosis has not been described in dogs (beyond an anecdotal case report, the PK-deficient Cairn Terrier[12]), there is no experience with iron chelation for any disease in dogs. The observed secondary hemosiderosis/-chromatosis in a PK-deficient Labrador Retriever suggests that chelation treatment may be warranted in this disease.

The newly identified and previously described mutations in different breeds simplify the diagnosis of PK deficiency in dogs and their knowledge is invaluable to breeders seeking to detect carriers. Erythrocytic PK deficiency should be considered in younger chronically anemic dogs, but the more cumbersome enzyme testing and search for novel mutations is still required to make a definitive diagnosis in those breeds whose mutations have not been identified. This erythroenzymopathy does not only cause hemolytic anemia but also a hemosiderosis, hemochromatosis or both secondary to chronic hemolysis and an as yet unexplained osteosclerosis.

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Abbreviations

Ala
: alanine

bp
: base pair

EDTA
: ethylenediaminetetraacetic acid

gDNA
: genomic DNA

Gln
: glutamine

Hb-O_2
: hemoglobin-oxygen

HCT
: hematocrit

H&E
: hematoxilen and eosin

HFE
: hemochromatosis gene

HGB
: hemoglobin

IMHA
: immune-mediated hemolytic anemia

Iso
: isoleucine

MCHC
: mean corpuscular hemoglobin concentration

MCV
: mean corpuscular volume

M-PK
: M-type pyruvate kinase protein

OF
: osmotic fragility

PCR
: polymerase chain reaction

PK-LR
: pyruvate kinase gene

R-PK
: R-type pyruvate kinase protein

Ser
: serine

SIFT
: sorting intolerant from tolerant

Val
: valine

Footnotes

[a]Qiagen: Germantown, MD

[b]Ambion: Austin, TX

[c]Applied Biosystems: Foster City, CA

[d]Invitrogen: Carlsbad, CA

[e]Takara: Otsu, Shiga, Japan

[f]DNASTAR, Madison, WI

[g]New England Biolabs: Ipswich, MA

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